

Evidence boundary and the shortest path to a defensible internal benchmark

A shared software platform is feasible; efficacy evidence must remain modality- and artifact-specific

	Software execution	H&E fold localization	Crack/tear localization	Natural COMET/ CosMx artifacts	Operational safety
Synthetic engineering smoke	Demonstrated	Not efficacy	Not efficacy	Not efficacy	Not measured
Public H&E locked test	Demonstrated	Exploratory	No labels	Out of scope	Not measured
Public multiplex proxy	Demonstrated	Out of scope	Proxy only	Proxy only	Not measured
Internal adjudicated locked test	Required next	Required next	Required next	Required next	Required next
Prospective silent validation	Required	Monitor	Monitor	Monitor	Required

Next decision gate: dual-reviewer ontology pilot → group-disjoint development/calibration → untouched internal test → prospective workflow validation.

A pooled three-modality score must not compensate for failure in any required modality, artifact subtype, or acquisition stratum.